Water Quality Monitoring and Pool Maintenance System

Centre of Excellence in Artificial Intelligence

(CEAI)

Group ID: R26-IT-143



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BSc (Hons) in Information Technology Specialized in Information Technology |

PROBLEM BACKGROUND

- Manual observation fails in high-density crowd scenarios.
- Contaminant build-up accelerates with swimmer volume.
- Fixed schedules lead to chemical waste or hygiene risks.
- Human error in monitoring leads to fatal drowning risks.
- Delayed response to garbage/debris intrusion impacts filter life.



STAKEHOLDERS & IMPACT



Pool Managers

Optimize operational costs and resource allocation for maintenance staff.



Lifeguards

Augmenting human vision with AI behavior analysis for rapid drowning detection.



Pool Users

Ensuring health safety via real-time water quality monitoring and hygiene alerts.



Maintenance

Switching from reactive to predictive maintenance scheduling based on usage data.

EXISTING LIMITATIONS

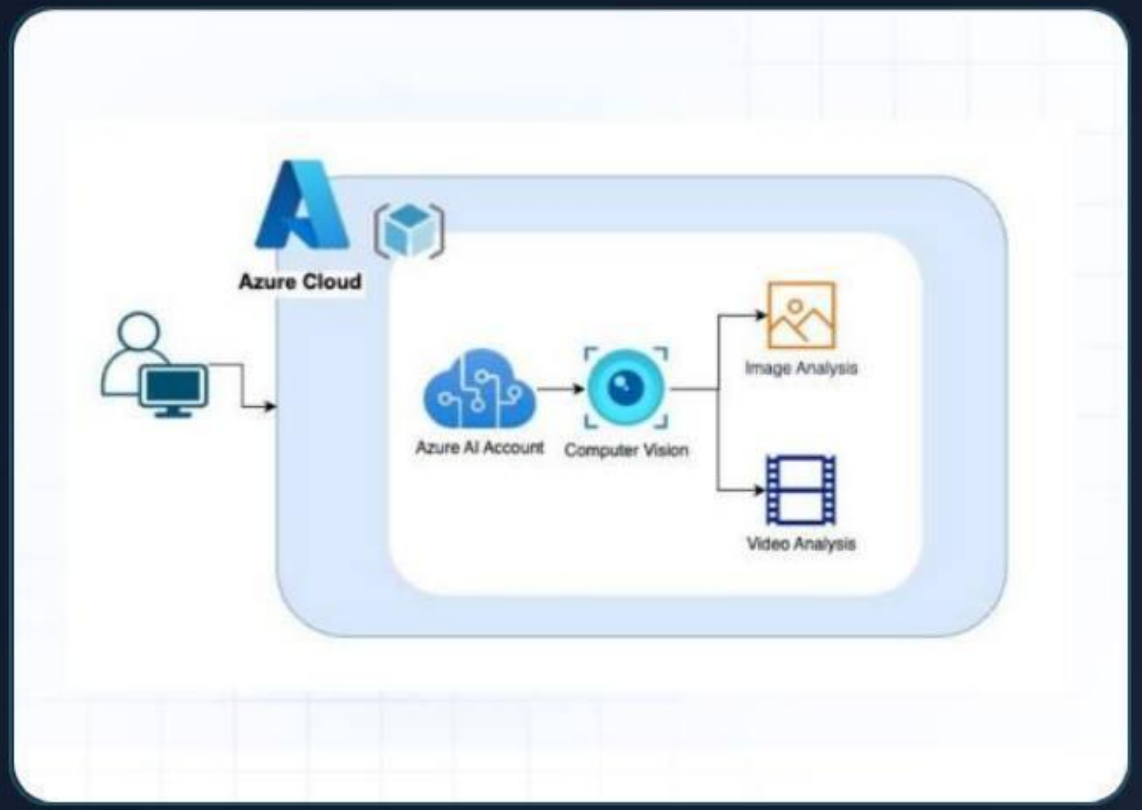
Current Methods

- Basic SCADA sensor monitoring (pH only)
- Fixed hourly lifeguard rotations
- Weekly manual chemical titration

The Gap

- Lack of real-time crowd density analysis
- No predictive modeling for water decay
- Delayed drowning behavior detection
- Fragmented, non-integrated monitoring

SYSTEM ARCHITECTURE



- **Data Layer:** CCTV Video Streams + IoT Sensor Array (pH, Chlorine, Temp).
- **Processing Layer:** Computer Vision (YOLO) + ML Prediction Models.
- **Decision Layer:** Adaptive Maintenance Scheduler.
- **Interface Layer:** Web-based Pool Management Dashboard.

TEAM CONTRIBUTIONS



Team Leader : M.H.M Saajith
Component : Crowd-Aware Maintenance Scheduling Module



Team Member : Y Nilogini
Component : Water Quality Classification and Prediction Module



Team Leader : N.M Shafan
Component : Swimmer Behavior and Drowning Detection Module



Team Leader M.Y.S Arsalan
Component : Garbage Intrusion Detection Module

COMPONENT 1: CROWD MAINTENANCE
IT22156310



Problem Definition

- ✓ Current systems **do not consider swimmer density** when planning maintenance.
- ✓ High swimmer density leads to rapid water quality degradation.



Research Objectives

- ✓ Calculate crowd density levels using CCTV video streams.
- ✓ Correlate swimmer density with water quality degradation patterns.

Novelty

- ✓ Calculate crowd density levels using CCTV video streams.
- ✓ Correlate swimmer density with water quality degradation patterns.
- ✓ Generate dynamic maintenance scheduling recommendations.

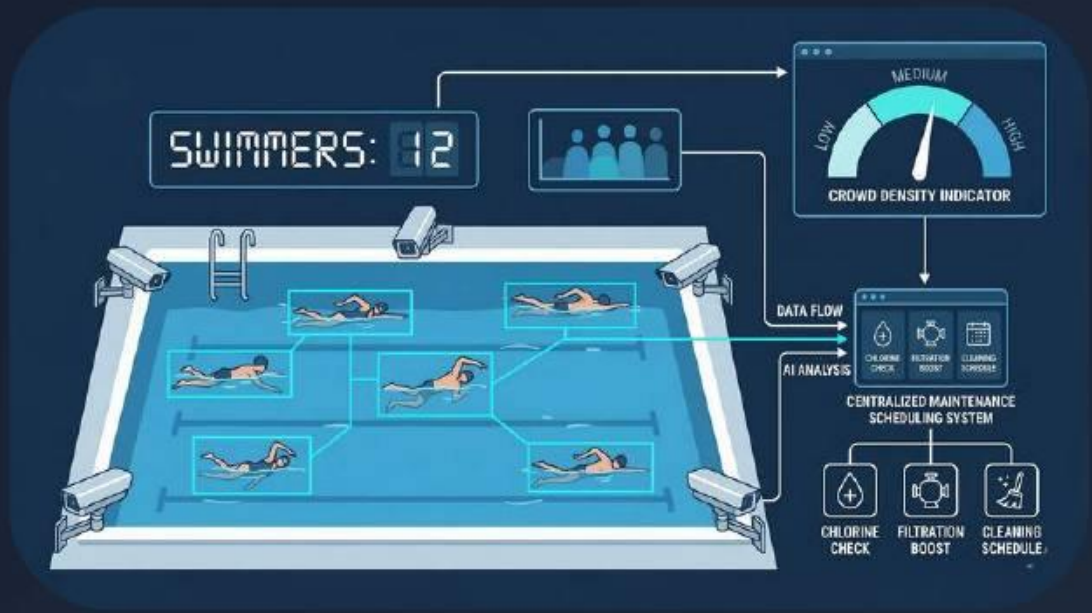
COMPONENT 1: CROWD MAINTENANCE
IT22156310

Methodology

- ✓ Capture real-time video using CCTV cameras installed around the pool.
- ✓ Use YOLO-based computer vision models to detect swimmers.
- ✓ Estimate crowd density based on swimmer count and pool area.
- ✓ Analyze crowd patterns to determine maintenance requirements.
- ✓ Send crowd density data to the maintenance scheduling module.

Technologies

- ❖ Computer Vision
- ❖ YOLO Object Detection
- ❖ Python
- ❖ OpenCV
- ❖ Data Analytics



COMPONENT 1: CROWD MAINTENANCE
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Validation

- ✓ Evaluate accuracy of swimmer detection model.
- ✓ Compare automated swimmer count with manual observations.
- ✓ Analyze effectiveness of crowd-based maintenance scheduling.

Data Availability

- ✓ CCTV video streams from swimming pool environment.
- ✓ Simulated datasets for swimmer detection.

Security and Ethics

- ✓ Ensure privacy protection of swimmers.
- ✓ Avoid storing personally identifiable information.
- ✓ Follow ethical guidelines for surveillance systems.

COMPONENT 1: CROWD MAINTENANCE
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System Overview

CCTV Cameras
→ Video Processing Module
→ Swimmer Detection
→ Crowd Density Estimation
→ Maintenance Scheduling Module

Dependencies

- ✓ Video processing module
- ✓ System database
- ✓ Pool management dashboard

COMPONENT 2: QUALITY PREDICTION
IT22247254



Problem Definition

- ✓ Water quality degradation often occurs suddenly due to high swimmer activity.
- ✓ Existing monitoring systems cannot predict water quality changes in advance.



Research Objectives

- ✓ Collect sensor data from water quality monitoring devices.
- ✓ Train machine learning models to predict water quality levels.
- ✓ Classify water quality into Safe, Warning, and Critical categories.

Novelty

- ✓ Introduces predictive water quality monitoring using machine learning.
- ✓ Identifies key parameters influencing water quality degradation.
- ✓ Enables proactive pool maintenance decisions.

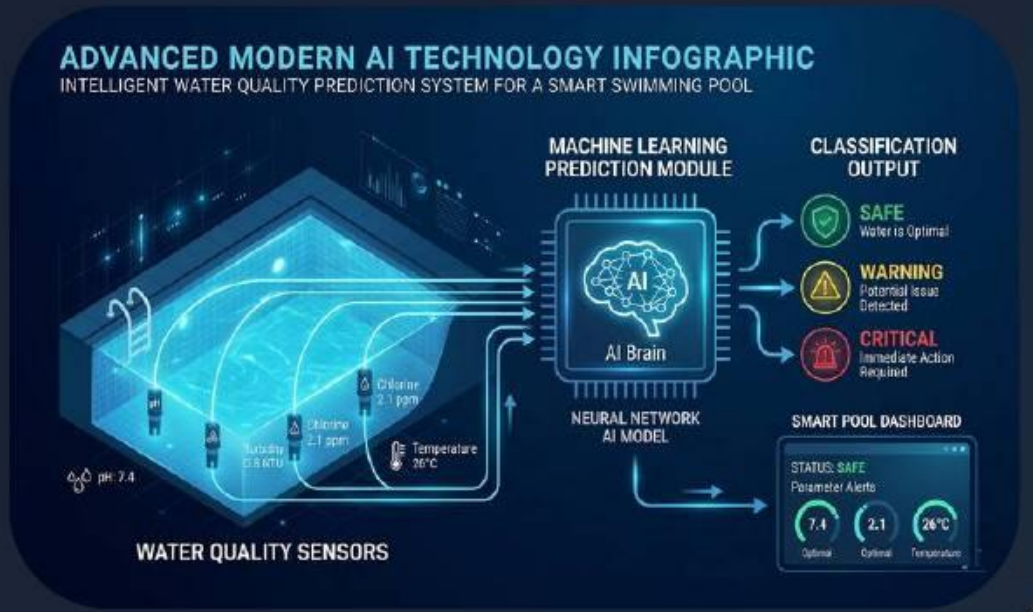
COMPONENT 2: QUALITY PREDICTION
IT22247254

Methodology

- ✓ Collect sensor data including:
  - pH
  - Turbidity
  - Chlorine
  - Temperature
- ✓ Perform data preprocessing and normalization.
- ✓ Train machine learning models for classification and prediction.
- ✓ Generate real-time water quality status alerts.

Technologies

- ❖ Machine Learning
- ❖ Python
- ❖ Data preprocessing
- ❖ Classification algorithms



COMPONENT 2: QUALITY PREDICTION
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Validation

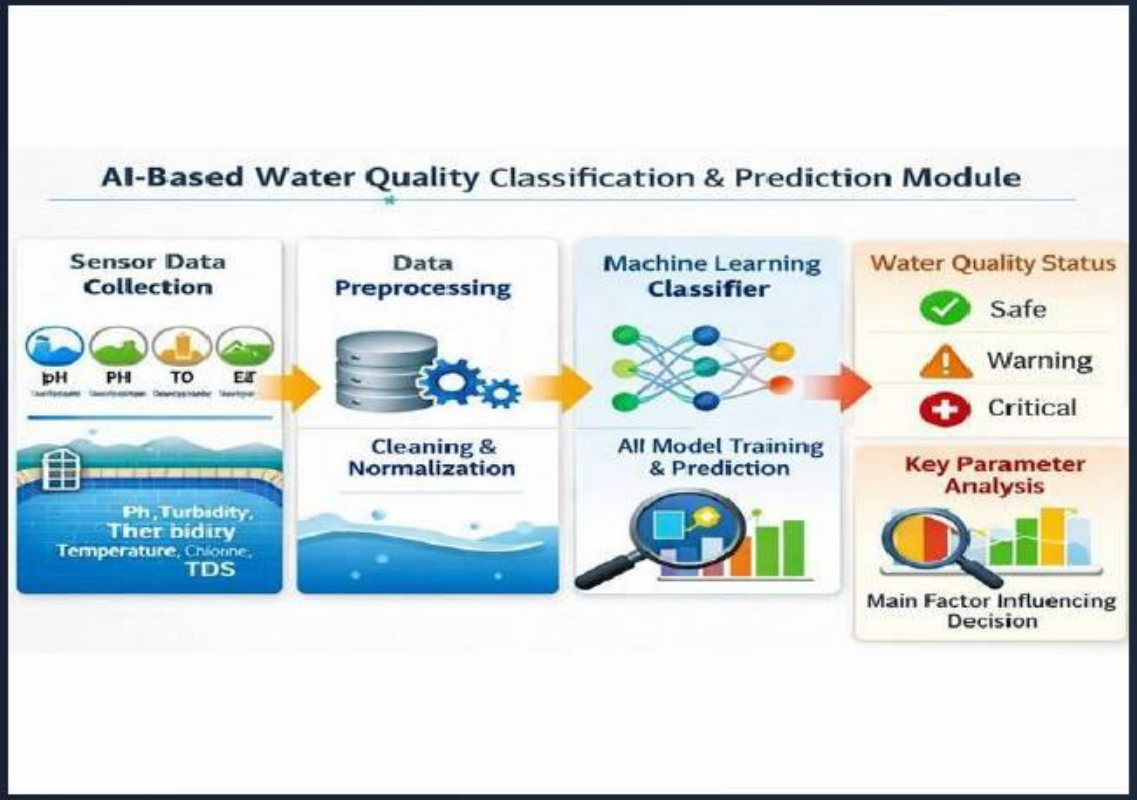
- ✓ Evaluate prediction performance using:
  - Accuracy
  - Precision
  - Recall
  - F1-score

Data Availability

- ✓ Water quality sensor datasets
- ✓ Public environmental monitoring datasets

Security and Ethics

- ✓ Secure storage of environmental monitoring data
- ✓ Ensure responsible use of collected sensor information



COMPONENT 2: QUALITY PREDICTION
IT22247254

System Overview

Water Quality Sensors
→ Data Collection
→ Data Preprocessing
→ Machine Learning Model
→ Water Quality Prediction

Dependencies

- ✓ Sensor module
- ✓ System database
- ✓ Pool management dashboard

COMPONENT 3: DROWNING DETECTION
IT22248480



Problem Definition

- ✓ In crowded pools it is difficult to monitor every swimmer simultaneously.
- ✓ Drowning incidents may not be detected immediately.



Research Objectives

- ✓ Detect swimmers using computer vision models.
- ✓ Analyze swimmer behavior and movement patterns.
- ✓ Identify abnormal behaviors indicating potential drowning risks.

Novelty

- ✓ Enables early drowning detection using computer vision.
- ✓ Enhances pool safety through automated alerts.

COMPONENT 3: DROWNING DETECTION
IT22248480

Methodology

- ✓ Capture real-time video streams from CCTV cameras.
- ✓ Apply deep learning models to detect swimmers.
- ✓ Track swimmer movements over time.
- ✓ Identify abnormal patterns such as:
  - prolonged inactivity
  - distress movements
  - sinking behavior

Technologies

- ❖ Computer Vision
- ❖ Deep Learning
- ❖ YOLO / CNN models
- ❖ Python



COMPONENT 3: DROWNING DETECTION
IT22248480

Validation

- ✓ Compare AI-based detection with manual observation.
- ✓ Measure detection accuracy and response time.

Data Availability

- ✓ CCTV video datasets
- ✓ Simulated drowning scenarios

Security and Ethics

- ✓ Protect privacy of swimmers.
- ✓ Ensure video data is used only for research purposes.

COMPONENT 3: DROWNING DETECTION
IT22248480

System Overview

Water Quality Sensors
→ Data Collection
→ Data Preprocessing
→ Machine Learning Model
→ Water Quality Prediction

Dependencies

- ✓ Sensor module
- ✓ System database
- ✓ Pool management dashboard

COMPONENT 4: GARBAGE DETECTION
IT22277336



Problem Definition

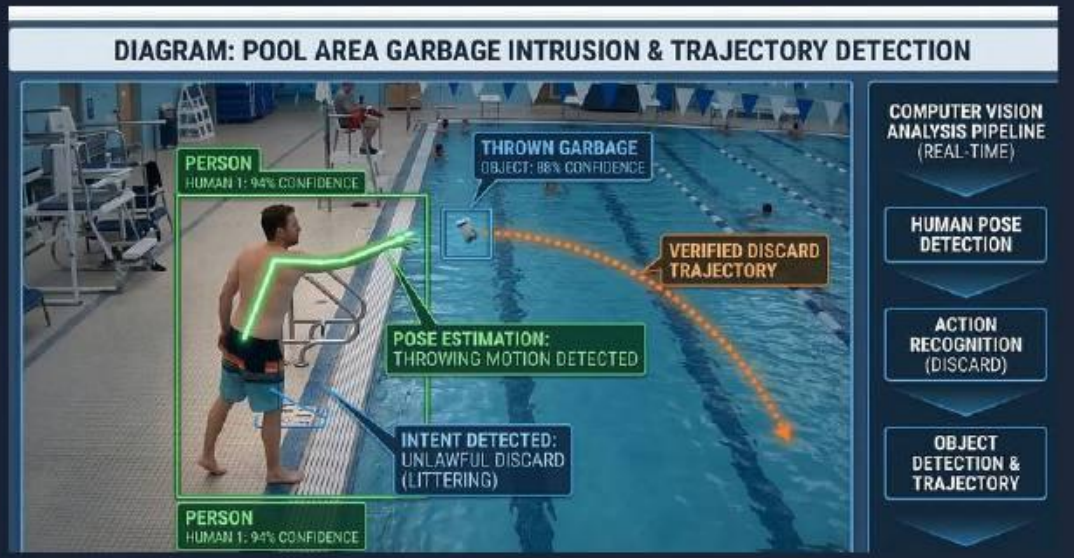
- ✓ Swimming pools require strict hygiene and safety monitoring
- ✓ Garbage such as plastic bottles, wrappers, and food waste may enter pools
- ✓ Current monitoring mainly relies on manual supervision or basic CCTV
- ✓ Manual monitoring may miss incidents or respond slowly
- ✓ Lack of intelligent systems that detect both objects and human behavior

Research Objectives

- ✓ Develop an AI-based monitoring system for swimming pool environments
- ✓ Detect garbage objects entering the pool using computer vision
- ✓ Analyze human actions near the pool area
- ✓ Identify intentional garbage throwing behavior

Novelty

- ✓ Introduces intent-aware garbage intrusion detection
- ✓ Combines object detection + human action recognition



COMPONENT 4: GARBAGE DETECTION
IT22277336

Methodology

- ✓ Capture real-time video streams using pool surveillance cameras
- ✓ Perform video preprocessing (frame extraction, noise reduction)
- ✓ Apply AI object detection model to identify garbage items
- ✓ Implement human action recognition module
- ✓ Combine object detection and behavior analysis for intent evaluation
- ✓ Generate real-time alerts and intrusion logs

Technologies

- ❖ Python programming language
- ❖ Computer Vision (OpenCV)
- ❖ Deep Learning Frameworks (TensorFlow / PyTorch)
- ❖ Object Detection Models (YOLO / CNN-based models)
- ❖ Database systems for storing intrusion logs
- ❖ Cloud or server hosting for system deployment

COMPONENT 4: GARBAGE DETECTION
IT22277336

Validation

- ✓ Evaluate object detection accuracy.
- ✓ Test system performance in different pool scenarios
- ✓ Compare detected events vs actual events.

Data Availability

- ✓ Collect sample pool environment videos for testing
- ✓ Generate annotated datasets for garbage objects

Security and Ethics

- ✓ Ensure privacy protection for individuals captured in video
- ✓ Use data only for research purposes
- ✓ Store video and logs in secure databases
- ✓ Avoid misuse of surveillance data

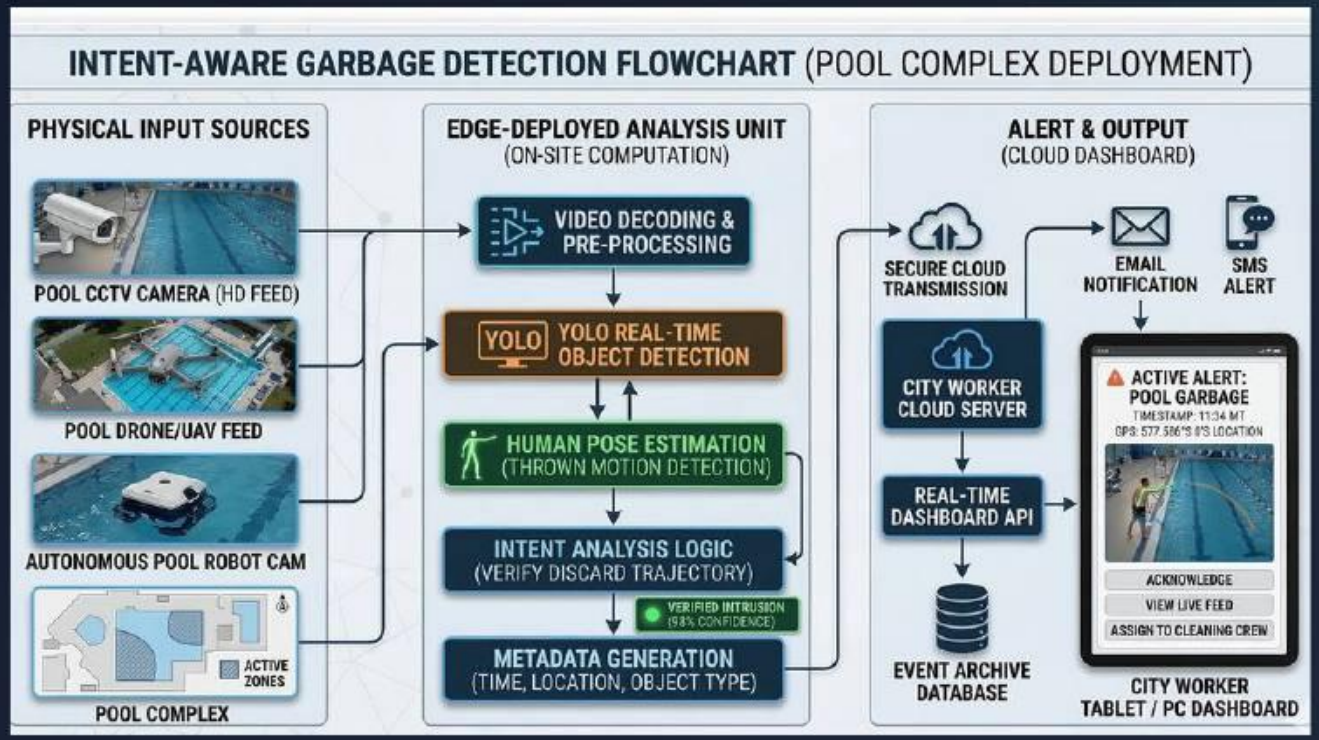
COMPONENT 4: GARBAGE DETECTION
IT22277336

System Overview

CCTV Cameras
→ Video Preprocessing Module
→ Garbage Object Detection (AI Model)
→ Human Action Analysis
→ Intent Detection Engine
→ Real-Time Alert System
→ Intrusion Log Database

Dependencies

- ✓ Video processing module
- ✓ AI object detection model
- ✓ Human action recognition module
- ✓ System database
- ✓ Alert notification system
- ✓ Pool management dashboard



VALIDATION & ETHICS



Validation

Accuracy, Precision, Recall metrics with 90%+ target for drowning detection.



Privacy

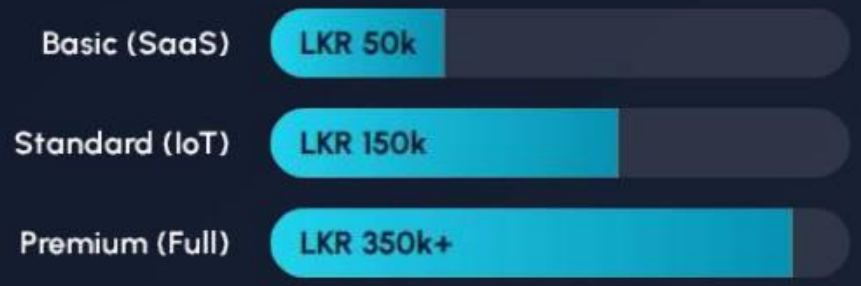
Real-time face blurring on CCTV streams to protect swimmer anonymity.



Security

AES-256 encryption for sensor data storage on the cloud.

MARKET & COMMERCIALIZATION



Target Markets

- Luxury Hotels & Resorts
- Municipal Sports Complexes
- Private Residential Clubs
- International Schools

TIMELINE & WBS



WBS: Project Lead → Research → ML Training → Hardware Integration → QA Testing

THANK YOU

Thank you for your attention.

We welcome any questions or feedback from the panel.

AI-Based Smart Swimming Pool Monitoring and Maintenance System
Final Year Research Project – BSc (Hons) Information Technology